

2 0 2 6 T E L E M E T R Y C A T A L O G U E

Power *OH*
Racing Electronics



Power On di Lorenzo Wohlgemuth

🏠 LOCALITÀ PIERA 28A - 38038 TESERO (ITALY)

☎ +39 0462 341015

🖨 +39 0462 248393

🌐 WWW.POWERON.IT

✉ INFO@POWERON.IT

📍 46,290254°N - 11,4917604°E

📘 POWER ON RACING ELECTRONICS

🌐 POWERON RACING ELECTRONICS

📷 POWERONRACINGELECTRONICS



In the early 50s, building a radio link meant to literally climb our Dolomites. Massive systems were installed on mountains where huge and heavy antennas arised. These pioneering systems appeared, the term "telemetry" had not been used yet. Only after first space missions, it became necessary to get precise real-time data through telemetry systems.

//MOST ADVANCED INTERNET TELEMETRY SYSTEM

With the advent of mobile telephones and Internet network, new communication systems were developed. 3G, 4G and 5G protocols made data roaming possible through very small appliances.

Power On has internally developed its own Internet telemetry: an innovative system which is compatible with all main Motorsport electronics brands. Our system does not require any ground station connected to the telephone network, but just a simple connection to the Internet. It has been tested worldwide and successfully used in main car endurance races.

Don't leave the success of your measurement projects to chance: rely on the know-how of measurement technology experts.

INDEX

3	//MOST ADVANCED INTERNET TELEMETRY SYSTEM
4	//FEATURES AND SERVICES - TS TELEMETRY FAMILY
7	TS124G
9	TS154G GYR
11	TS155G LANNER
12	//WHAT'S INCLUDED IN THE KIT
13	//POWER ON SERVERS
14	//ANTENNAS
16	//SOFTWARE & LICENSES
17	//WIRING LOOMS
18	//PARTNERSHIP
19	//COMPATIBILITY
20	//TECHNICAL LAYOUT
21	//TECHNICAL FEATURES
22	//NOT ONLY TELEMETRY
23	//REPRESENTED BRANDS

//FEATURES AND SERVICES - TS TELEMETRY FAMILY

THE HARDWARE

Small, light, and suitable for working in one of the toughest and most difficult environments: a racing car's cockpit.

CAR CONNECTION

Connection to the car system is obtained via RS232 or Ethernet, whereas most Motorsport devices normally provide telemetry data transmission through serial link only. By replacing traditional RS232 with a direct Ethernet link between the modem and the car, you achieve much higher data rates and faster, more reliable telemetry.

CLOUD

System's core is represented by a significant use of the Internet cloud. Our servers guarantee reliable real time communication between cars on track and garages, headquarters, or any other place around the world. Power On manages a network of servers spread over different continents to ensure fast and stable connection in North America, Europe and Asia. The system autonomously recognizes car location and

automatically chooses the most appropriate server. Our TS155G Falcon Lanner version uses a network of dedicated servers.

BSF™ PROTOCOL

The BSF™ (Best Signal Forwarding) protocol, developed by Power On, continuously analyzes the quality of the data received and automatically forwards the best available to Power On Bridge. This process minimizes the number of track sectors that remain out of signal coverage.

ETHERNET SWITCH

The integrated Ethernet switch enables pass-through umbilical connectivity for vehicle controllers equipped with a single Ethernet port. This way, telemetry can be perfectly integrated into the network layout of the car.

MULTI STREAM DATA

The Multi Stream Data function is available only with compatible data acquisition systems and can be used through both Ethernet and RS232 connections. This feature enables data to be distributed across different team

departments (e.g. team data, engine data, and OEM data) and also allows each stream to be managed with dedicated license sets (main + mirror).

GROUND RECEIVER

Our telemetry doesn't need any ground receiver, but only a reliable Internet connection. It is possible to access car data from two different locations (or more, if further licenses are activated).

GARAGE NETWORK

At each ground connection, a network with potentially unlimited number of PCs can be created. Any authorized user will then be able to access data of all cars of a particular team. At the same time, data may also be available to another PC network, such as team headquarters, where other technicians may be authorized to use them.

PROGRAMMING

All telemetry programming phases are performed via a dedicated ethernet connection. This connection can also be accessed for device diagnostics.

POWER ON BRIDGE

The management of our telemetries takes place through our dedicated Power On Bridge software.

This software acts as the link between our servers and the ground-based data reception. It also serves as the tool that enables sharing of telemetry data with end-user's data analysis system.

UPDATES

Our Power On Bridge and servers are constantly updated: any new specification of Internet providers is monitored to offer more powerful products. On our website, we regularly publish updated firmwares with improved functionalities. Telemetry firmware can be upgraded autonomously or through our support service.

DATA SECURITY

Your car data is protected by our protocol, as well as by a personal license. The state-of-the-art data encryption ensures a full protection of all car information.

INTERNET NETWORK

Our telemetry system does not require any specific data SIM, so you can autonomously choose your favourite mobile service provider. You shall only remember to remove PIN from the SIM and set the appropriate APN.

Telemetry functionality is guaranteed with any provider, even with unbranded data SIMs purchased at any airport.

Nevertheless, we provide a SIM of CObyEX, our official communication service partner. To use this SIM you only need to directly contact CoByEx; all contact information will be available on telemetry's accompanying documentation.

TECHNICAL SUPPORT

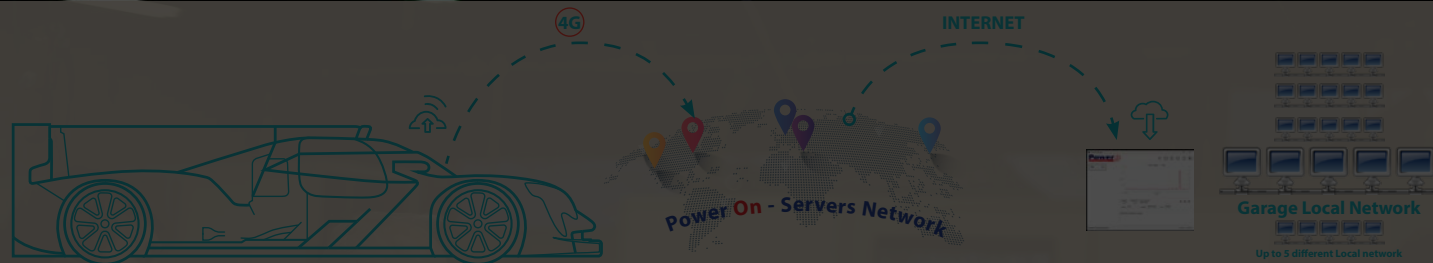
While using our telemetry, you will be fully supported by our technical team. If our manual should not be clear enough, you can get our remote support to correctly set up of your data acquisition software to allow an easy use regardless of the brand.

While running, we constantly monitor the status of our servers. If required, we can

support you throughout the entire race to ensure a seamless functionality of the telemetry service.



FEATURES AND SERVICES



LINK TO CAR

Link to your car system via RS232.



GROUND NETWORK

Thanks to the ground networks, an unlimited number of PCs can be connected to each other and make car data available to all authorized members of the team.



PROGRAMMING

Programmable via WiFi, with no cable required.



SINGLE SIM

Choose your operator, remove the PIN and go!



CLOUD

Our Servers guarantees reliable real-time communication.



UPGRADES

We regularly publish updated firmwares with improved functionalities. Telemetry firmware can be upgraded autonomously or through our support service.



DATA SECURITY

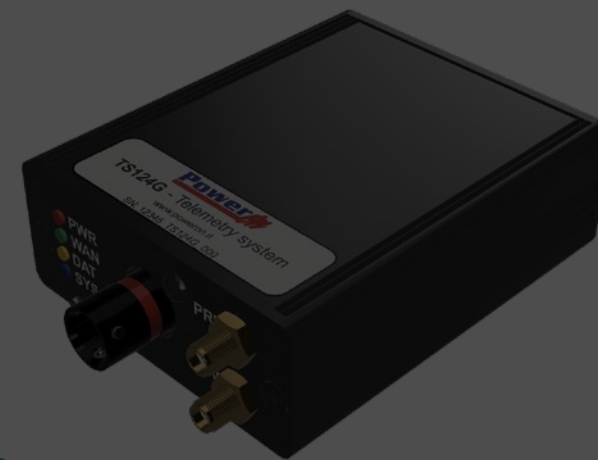
State-of-art encryption and personal licence ensures a full protection of all car information.

TS124G

4G TELEMETRY KIT

P002006101 - TS 124G

4G TELEMETRY DEVELOPED BY POWER ON. COMPATIBLE WITH THE MOST IMPORTANT MOTORSPORT ELECTRONIC BRANDS



DETAILS

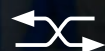
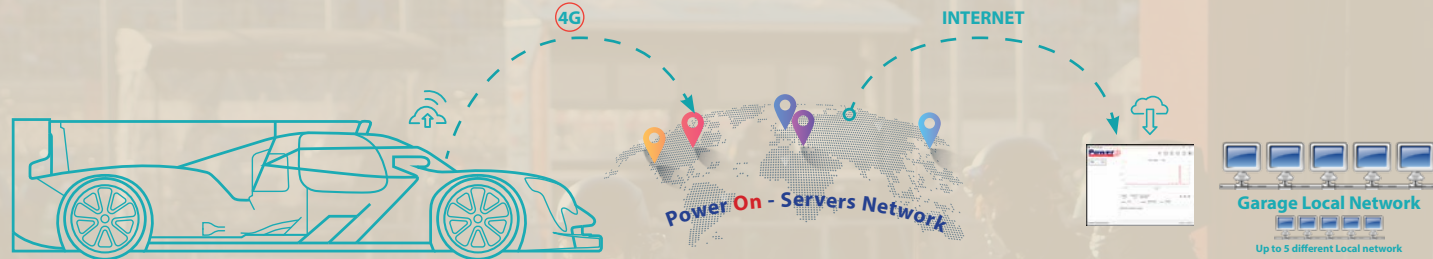
- 4G SIM card of any mobile service providers
- Programming via Wi-Fi
- RS232 data stream
- 1x Main Licence + 1x Mirror Licence included
- Multiple Licence possibility*: receive data all over the world
- One hour free assistance for the first configuration



*EXTRA MIRROR LICENCES AVAILABLE

NO LONGER AVAILABLE

FEATURES AND SERVICES



LINK TO CAR

Link to your car system via RS232 or Ethernet.



GROUND NETWORK

Thanks to the ground networks, an unlimited number of PCs can be connected to each other and make car data available to all authorized members of the team.



POWER ON BRIDGE

Dedicated software for seamless telemetry management.



UPGRADES

We regularly publish updated firmwares with improved functionalities. Telemetry firmware can be upgraded autonomously or through our support service.



PROGRAMMING

Dedicate Ethernet port.



SINGLE SIM

Choose your operator, remove the PIN and go!



CLOUD

Our Servers guarantees reliable real-time communication.



DATA SECURITY

State-of-art encryption and personal licence ensures a full protection of all car information.

TS154G GYR

4G TELEMETRY KIT



*GET THE MAXIMUM DATA RATE FROM YOUR CAR

**SEPARATE DATA STREAM FROM THE SAME TELEMETRY MODULE. *DOUBLE DATA, SAME WEIGHT!*

***EXTRA MIRROR LICENCES AVAILABLE

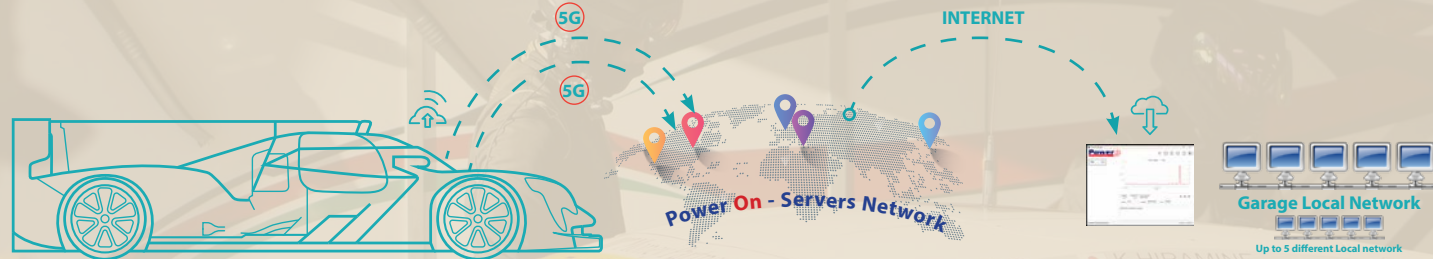
P002010100 - TS154G FALCON - KIT GYR

4G TELEMETRY DEVELOPED BY POWER ON. COMPATIBLE WITH THE MOST IMPORTANT MOTORSPORT ELECTRONIC BRANDS

DETAILS

- 4G SIM card of any mobile service providers
- Substitute the old TS124G PowerOn Telemetry
- RS232 data stream
- Ethernet data stream upgrade available*
- Programming via Ethernet dedicated port
- Ethernet switch functionality
- Multiple data stream upgrade available**
- 1x Main Licence + 1x Mirror Licence included
- Multiple Licence possibility***: receive data all over the world
- One hour free assistance for the first configuration

FEATURES AND SERVICES



LINK TO CAR

Link to your car system via RS232 or Ethernet.



DATA SECURITY

State-of-art encryption and personal licence ensures a full protection of all car information.



GROUND NETWORK

Thanks to the ground networks, an unlimited number of PCs can be connected to each other and make car data available to all authorized members of the team.



POWER ON BRIDGE

Dedicated software for seamless telemetry management.



UPGRADES

We regularly publish updated firmwares with improved functionalities. Telemetry firmware can be upgraded autonomously or through our support service.



BSF

Power On developed: always the best data stream from your car.



DUAL SIM

Dual SIM, Double Modem: Always-on connectivity.



CLOUD

Our Servers guarantees reliable real-time communication powered by BSF™.

TS155G LANNER

DUAL SIM 5G TELEMETRY KIT

P002010450 - TS155G FALCON - KIT LANNER

DUAL SIM 5G TELEMETRY DEVELOPED BY POWER ON.
COMPATIBLE WITH THE MOST IMPORTANT MOTORSPORT
ELECTRONIC BRANDS



*SEPARATE DATA STREAM FROM THE SAME TELEMETRY MODULE. *DOUBLE DATA, SAME WEIGHT!*

**EXTRA MIRROR LICENCES AVAILABLE

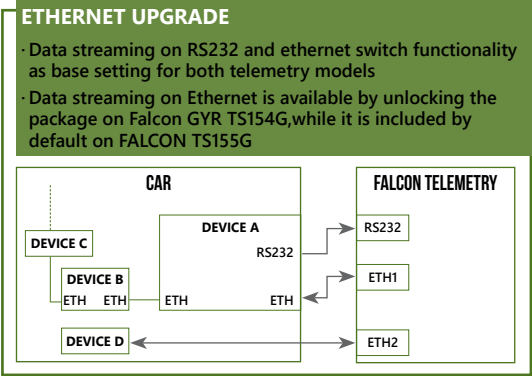
DETAILS

- Dual SIM 5G of any mobile service providers
- BSF™ protocol (for the best data transmission)
- Ethernet data stream included
- Programming via Ethernet dedicated port
- Ethernet switch functionality
- Multiple data stream upgrade available*
- Multiple Licence possibility**: receive data all over the world
- One hour free assistance for the first configuration

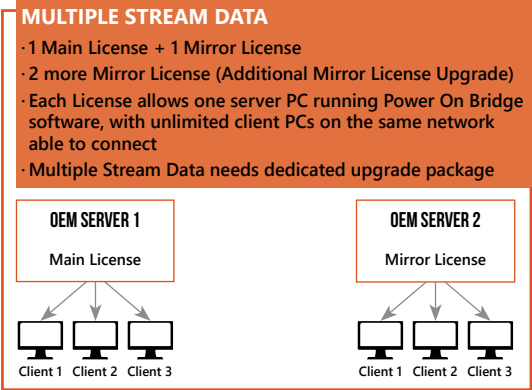
//WHAT'S INCLUDED IN THE KIT

	 P 002 010 100 TS154G Falcon - Kit Gyr	 P 002 010 450 TS155G Falcon - Kit Lanner
Telemetry model	TS154G	TS155G
Magnetic Antenna	✓	✓
SIM card adapter	✓	✓
Software license	✓ (first year)	✓ (first year)
Tech support	✓	✓
RS232 stream	✓	✓
Ethernet stream	✗ (to be ordered separately)	✓
Sim Card slot	1 x 4G	2 x 5G
Telemetry loom	✗ (to be ordered separately)	✗ (to be ordered separately)

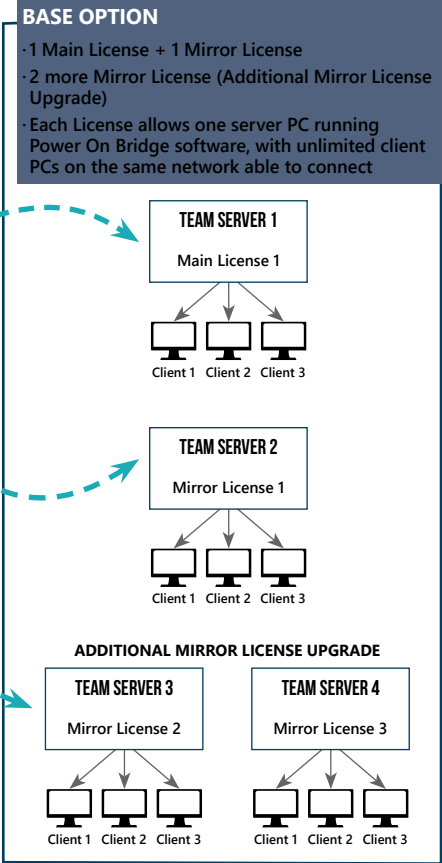
CAR DATA STREAM



EXTRA DATA STREAM



MAIN DATA STREAM



ACCESSORIES

ANTENNAS / LICENSES / WIRING LOOMS / SPARE PARTS

OUR TELEMETRY IS SUPPLIED AS A READY-TO-INSTALL KIT FOR IMMEDIATE USE.
ADDITIONAL ACCESSORIES CAN BE ADDED TO TAILOR THE SYSTEM PERFECTLY TO YOUR NEEDS

//ANTENNAS

APPROVED ANTENNAS TO BE USED WITH OUR TELEMETRY FOR OPTIMAL PERFORMANCE



MAGNETIC 4G ANTENNA

Magnetic stylus antenna with SMA connector and 3 metres cable.

DETAILS

- 4G 699-960 MHz
- LTE 1710-2700 MHz
- Impedance 50 Ohms
- Vertical polarization
- Max gain: 2.2 dBi

P 002 003 020



ADHESIVE 4G ANTENNA

Adhesive antenna for roof or windscreen with SMA connector and 2 metres cable.

DETAILS

- 4G 699-960 MHz
- LTE 1710-2700 MHz
- Impedance 50 Ohms
- Linear polarization
- Max gain: 2 dBi

P 002 003 040



ADHESIVE 5G ANTENNA

Adhesive antenna for roof or windscreen with SMA connector and 2 metres cable.

DETAILS

- 4G 699-960 MHz
- LTE 1710-2700 MHz
- 5G 3300-4900 MHz
- Linear polarization
- Max gain: 2 dBi

P 002 003 050



MAGNETIC/ADHESIVE 2X5G ANTENNA

2x5G MIMO Adhesive/Magnetic Mount Puck Antenna with 2xSMA connectors and 2 metres cables.

DETAILS

- 4G/5G/LTE 600-6000MHz
- Linear polarization
- Max. gain: 2 dBi
- IP67 WaterProof

P 002 003 090

//SOFTWARE & LICENSES

MANAGEMENT SOFTWARE AND LICENSES

POWER ON BRIDGE

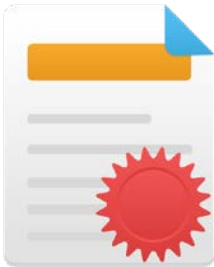
Our telemetry is managed through our Power On Bridge software, which ensures a secure and seamless connection between your laptop and our server, giving you reliable access to your telemetry data at all times, anywhere in part of the world.



DETAILS

- Compatible with Windows
- Free software

[HTTPS://T1.POWERONTELEMETRY.IT](https://t1.powerontelemetry.it)



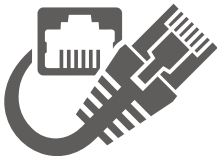
SOFTWARE LICENSE

Annual Licence allows you to keep using our server for telemetry data stream.

DETAILS

- One year validity
- First year included

P 002 010 920



ETHERNET UPGRADE

Ethernet upgrade allows significantly higher data rates than traditional RS232 telemetry and increases connectivity options with a multiple data stream.

DETAILS

- One-time software unlock

P 002 010 901



ADDITIONAL MIRROR LICENSE

Additional mirror licenses are added to the mirror license included in the telemetry kit.

DETAILS

- Up to maximum 3 mirror licenses can be obtained

P 002 010 930



MULTIPLE DATA STREAM LICENSE

Possibility to split different data streams among different departments of the team (e.g., Team data, Engine data, OEM data).

DETAILS

- Different license sets (main + mirror) for each data stream
- Valid for one year, after it a dedicated renewal is required (P 002 010 920)

P 002 010 940

//WIRING LOOMS

UNIVERSAL LOOM (CUSTOM ON REQUEST)



UNIVERSAL RS232 LOOM

Universal wiring harness for all car brands. Supplied without any car interface connector.

DETAILS

- Length: 500 mm
- Supplied with free wires

A 24C 177 001-1

//PARTNERSHIP



Our telemetry system does not require any specific data SIM, so you can autonomously choose it. Our telemetry system works with any standard data SIM, allowing you to freely choose among your preferred mobile service providers. You shall only remember to remove the PIN and select the new operator.

Telemetry functionality is guaranteed with any provider, even with unbranded data SIM purchased at any airport. Nevertheless, we provide a SIM of **CObyEX**, our official communication partner for connection service.

COBYEX

Since its founding, **CObyEX** has served a wide range of industries, providing **customized mobile solutions** while continuously enhancing its customer service.

Lorenzo Wohlgemuth, Chief of **Power On Racing Electronics**, commented:

"In the past, our teams had to buy different SIM cards for each race abroad. Today, CObyEX's solution is perfect: a single SIM card works worldwide, no matter where the race takes place. This means our teams no longer need to worry about sourcing SIM cards when travelling around the world".

Darren Andrew, Director of **CObyEX**, added:

"We are thrilled to partner with Power On Racing Electronics. This cooperation allows us to demonstrate how simple and convenient our SIM solution is, alongside the high-quality service and support we provide to users. We hope racing teams will fully enjoy the benefits our SIM offers".

//COMPATIBILITY



Bosch Motorsport
Authorized Dealer

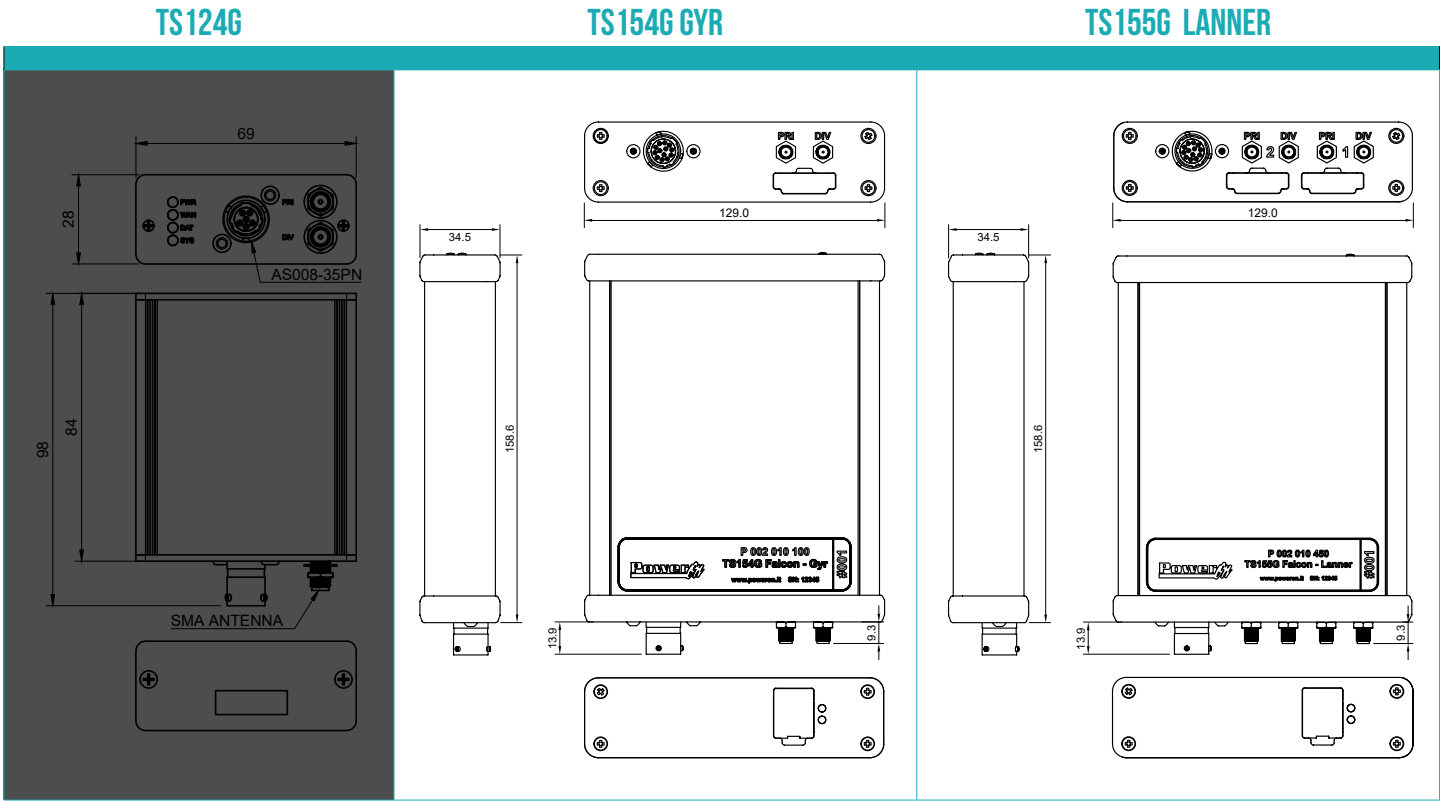
COSWORTH



... and many more



//TECHNICAL LAYOUT



//TECHNICAL FEATURES

TS124G		TS154G GYR		TS155G LANNER	
MECHANICAL DATA					
SIZE	99 x 69 x 28	172.5 x 129 x 34.5			MM
WEIGHT	155	185		205	G
TEMPERATURE RANGE	-20 TO 85°C	-40 TO 75°C (TESTED)			°C
ELECTRICAL DATA					
INPUT VOLTAGE	9-16	9-16			VOLT
MAX POWER CONSUMPTION	0,5	0,4 (@ 12V)		0,7 (@ 12V)	AMPERE
CONNECTORS	AS008-35PN	AS010-35PN			TYPE
ANTENNA CONNECTORS	2	2		4	NR
VISUAL CONTROLS	4	2			LED
DATA INTERFACES					
TYPE (1)	1 x RS232	1 x RS232 (UP TO 115200 BPS)			
TYPE (2)	/	2 x ETHERNET (100 MBPS)			
SIM CARD					
SLOT	1	1		2	NR
TYPE	MINI-SIM	MINI-SIM		MINI-SIM	
INTERNET COVERAGE & FREQUENCY					
5G	/	/	600 / 700 / 850 / 900 / 1800 / 2100 / 2600 / 3500		MHZ
4G/LTE	800/850/900/1800/2100/2600	800/850/900/1800/2100/2600			MHZ
3G/UMTS	850/900/2100	850/900/2100			MHZ
GSM	850/900/1800/1900	850/900/1800/1900			MHZ
COVERAGE	GLOBAL	GLOBAL			

In the interest of continuous product improvement, we reserve the right to alter without prior notice the specifications and features described in this catalog

//NOT ONLY TELEMETRY

Power On is a small company operating primarily in the motorsport electronics sector.

Lorenzo Wohlgemuth, the owner and founder, started the business after significant experience in Formula One, establishing a freelance operation that provides electronic support to major factory teams in both rally and circuit racing.

Back then, he undertook commercial and professional relations with the main Motorsport electronic manufacturers. Over the years, these business relationships have led him to become the Italian distributor for many products.

That's how **Power On** was born: a mixture of commercial support, technical competence and design skills. All manufactured components, including wiring looms and PCBs, were produced through outsourced technologies.

The financial crisis of 2008 has meant a revolution in Motorsport regulations, and therefore, **Power On** needed to evolve and set its own production. Nowadays, the company has 21 employees and cooperates with many Italian universities, as well as research centres, such as the nearby 'Trentino sviluppo' with its 400 and more researchers. It represents, in fact, an electronic company which is able to totally satisfy Motorsport's OEM needs and requests.

Finally, **Power On** is following many projects in the industrial sector as well, using the know-how and skills acquired in years of activity in the Motorsport.

Power
Racing Electronics *ON*

//REPRESENTED BRANDS

texense
sensors by TEXYS

 **BOSCH**

Bosch Motorsport
Authorized Dealer

PEAK
BY HMS NETWORKS

HellermannTyton

RACELOGIC

CORALBA
RALLY

 **AUMOVIO**
Engineering
Solutions

megatec
electronic

Ixxat
BY HMS NETWORKS



WHY LOOK ELSEWHERE?

PIERA 28A - 38038 TESERO (TN) - ITALY INFO@POWERON.IT - WWW.POWERON.IT
VAT IT 01535800229 - TEL. 0039 0462 341015 - FAX 0039 0462 248393

CAT1.4_042026